Hemant Pathak

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Education

Indian Institute of Technology (ISM) Dhanbad *Dhanbad, India*
Integrated M.Tech in Mathematics and Computing | GPA: **8.90 / 10.00** **2023 – 2028**

Relevant Coursework

Linear Algebra • Statistical Inference • Object-Oriented Programming • Operating Systems • Theory of Computation • GPU Computing • Time Series Analysis • Stanford CS231n: Computer Vision • Stanford CME295: Transformers & LLMs

Experience

Apparent Energy Solutions Pvt. Ltd. | Certificate **Apr 2026 – Present**
Software Engineer Intern — Agentic MCP & Retrieval Systems *Remote*

- Architected and deployed a **FastMCP server with 16+ autonomous tools**, enabling multi-step agentic workflows for business users and reducing token overhead by ∼40% via one-shot query bundling.
- Engineered a schema-aware retrieval pipeline (`describe_table` → `query_database`) with deterministic fallback paths, 30s timeout guards, and a 48-case test suite to eliminate SQL hallucinations.

Apparent Energy Solutions Pvt. Ltd. | Certificate **Jan 2026 – Mar 2026**
Software Engineer Intern — AI & Computer Vision *Remote*

- Orchestrated high-throughput CV pipelines on Google Street View and multi-spectral satellite imagery using a TensorRT-optimized **YOLOv11** model for pole detection and hybrid wire segmentation.
- Applied graph-theoretic algorithms with **GeoPandas** and **PostGIS** to model pole-to-pole connectivity, establishing a geospatial digital twin of India's electrical grid topology.

Aeroqube Inc. | Certificate **Aug 2025 – Dec 2025**
Software Engineering Intern (AI) *Remote*

- Built autonomous reasoning agents using **LangChain**, **LangGraph**, and **Gemini (Med Gemma 27B)** with a FAISS-backed RAG pipeline to automate clinical information synthesis and summaries.
- Fine-tuned **MED-SAM 2** on volumetric CT/X-ray datasets for zero-shot lesion segmentation (>90% Dice coefficient) and deployed containerized inference pipelines on **AWS SageMaker** via Docker.

Projects

Shell.ai Fuel Blend AI Challenge 2025 | *Machine Learning* | **Rank 82 / 7,020** (*Team: That's It*) Leaderboard

- Built a multi-target regression pipeline predicting 10 fuel-blend properties from 55 inputs; engineered interaction and heterogeneity features to capture non-linear mixing behaviour.
- Benchmarked **LightGBM**, **CatBoost**, **XGBoost**, Random Forest, and linear models with 5-fold cross-validation; combined them via hybrid ensembling and target-level optimization.

IntelliDesk AI — LLM-Powered Support Automation | *Python, Pinecone, Groq*

- Built an email-support workflow that classifies messages by category and priority and generates context-aware replies with **LLaMA-3.1-8B** via **Groq**.
- Implemented semantic retrieval with **Pinecone** and **SentenceTransformers** and structured JSON outputs for reliable downstream processing.

Technical Skills

Languages:	Python (async, typed), C++, Java, C, MySQL, PostgreSQL
AI/ML Stack:	TensorFlow, PyTorch, Scikit-Learn, LightGBM, CatBoost, XGBoost, LangChain, LangGraph, OpenCV, FastAPI, FAISS, Pinecone
Core Concepts:	Agentic Workflow Design & Deployment, LLM Orchestration, RAG, Embeddings, Memory Systems, API Integration & Webhooks, Data Pipeline Architecture, Human-in-the-Loop Systems, Autonomous Agent Frameworks, Computer Vision
Tools/Cloud:	AWS SageMaker, Azure Doc Intelligence, Docker, GitHub Actions CI, Git, Google Cloud, Groq, n8n, Zapier

Achievements

- **Runner-up**, Algoverse algorithmic trading hackathon; statistical arbitrage strategy outperformed 50+ teams.

Positions of Responsibility

- **Student Placement Coordinator**, Career Development Centre, IIT (ISM) Dhanbad
- **Member**, Fintech Club, IIT (ISM) Dhanbad — *investment markets, M&A deal analysis, and financial modelling*